Week 10 – Workshop
Geothermics: Finite Differences

Geophysics 3A: Potential Fields & Geothermics

July 23, 2026

Duration: 3 hours **Setting:** Workshop room

Preparation

Pre-reading: Course Notes, Ch. 18
Lecture videos:
Notes:

Purpose

This workshop builds intuition for finite difference methods through physical reasoning and interactive activities. Students first derive the heat equation from energy conservation, then experience how numerical models work by acting as nodes that update temperature based on their neighbours. The workshop then explores how boundary conditions represent geological assumptions and how numerical stability reflects limits on how heat can physically propagate. By the end, students should understand that finite difference methods are simply local energy balance, that heat flow emerges from neighbour interactions, and that both boundary conditions and stability have direct physical meaning.

Learning Outcomes

Students will:

- Understand finite difference equations as local statements of energy conservation.
- Develop intuition for how numerical solutions evolve through neighbour interactions.
- Interpret boundary conditions as geological assumptions.
- Understand numerical stability as a limit on physical information transfer.

Session Flow (220 min)

0–10 min: Warm-up — What breaks down?

Ask students to think back over the analytic solutions they have encountered in Weeks 8 and 9 (Fourier's law steady-state geotherm, half-space cooling, plate cooling). In pairs: *Name one geological situation where each analytic solution would fail. What assumption breaks down?* Collect responses to motivate numerical approaches as the natural next step.

10–25 min: Mini-lecture — From energy conservation to finite differences

- The heat equation as a statement of local energy balance on a small rock volume.
- Discretisation: replacing continuous derivatives with differences between neighbouring nodes.
- The finite difference stencil: each node updates based on its neighbours.
- Stability: why the time step cannot be arbitrarily large ($\Delta t < \Delta x^2/2\kappa$).

25–55 min: Activity 10.A: From Conservation of Energy to Finite Differences

Concept
Finite difference equations arise from applying conservation of energy to small rock volumes.

- Nodes represent local rock volumes.
- Heat flux enters and leaves each node.
- Heat production adds energy locally.

The equation is *not a numerical trick*. The Laplacian as a measure of curvature (difference from neighbours). This is just bookkeeping of energy.

55–90 min: Activity 10.B: Finite Differences as a Physical Process: Human Finite Difference Solver

Concept
Numerical models evolve through local interactions between neighbouring nodes.

Discuss:

- smoothing of temperature
- spreading of heat
- role of gradients

90–110 min: Activity 10.C: Boundary Conditions as Geological Assumptions

Concept: Boundary conditions determine what physical system the model represents.

- Fixed temperature (Dirichlet)
- Fixed heat flux (Neumann)

110–120 min: Break

120–145 min: When Analytic Solutions Break Down

Concept

- Analytic solutions require simplifications:
  - constant properties
  - simple geometries
  - simple boundary conditions
- Real geology violates these assumptions.

Discussion prompts

- What makes geological systems difficult to solve analytically?
- Why do irregular boundaries force numerical approaches?

Your Iberia exercise on the problem has several issues that make it impossible to develop an analytic form, including variable sediment thickness, lateral structure, varying boundary condition.

145–185 min: Activity 10.D: Numerical Stability and Information Transfer

Concept Stability limits impose a "speed limit" on thermal information:

$$\Delta t < \frac{\Delta x^2}{2\kappa}$$

- Heat propagates gradually through diffusion.
- Numerical schemes must respect this.
- small vs large time steps
- what happens when updates are too large
- stability = causality
- information cannot skip nodes

Instability is not just numerical—it is physically impossible behaviour.

185–205 min: Wrap-up

Key connections

- Finite differences = local energy balance + neighbour interaction
- Boundary conditions = geological assumptions
- Stability = physical constraint on information transfer

Discussion

- How does the numerical method compare to analytic solutions?
- What advantages do numerical models provide?
- Where might these methods be applied beyond heat flow?

205–220 min: Course Conclusion

This is the final workshop of the course. Rather than a preview, use this time to connect ideas across all ten weeks.

- **Fields and gradients** (Week 1): all geophysical observations are measurements of fields; gradients tell us where physical property contrasts occur.
- **Physical properties** (Week 2): the link between measurement and geology lives in physical properties and their mixing laws.
- **Inversion** (Week 3): going from data to model requires confronting non-linearity, non-uniqueness, and the role of prior information.
- **Gravity and isostasy** (Weeks 4–5): density contrasts at depth shape both the gravity field and the long-term mechanical balance of the lithosphere.
- **Magnetics and Fourier methods** (Weeks 6–7): vector fields, polarity, depth–wavelength relationships, and spectral filtering.
- **Geothermics** (Weeks 8–9): heat flow connects surface measurements to deep thermal structure via conduction, diffusion, and phase changes.
- **Finite differences** (Week 10): numerical methods let us move beyond analytic limits to model the real, heterogeneous Earth.

Ask students: *Which concept or technique from this course do you think will be most useful in your future work, and why?*

Workshop Notes

Workshop Notes

Activity 10.A
From Conservation of Energy to Finite Differences

Purpose

To understand finite difference heat flow equations as direct consequences of energy conservation applied to small volumes of rock.

Learning Outcomes

By the end of this activity, you should be able to:

- Derive the finite difference heat equation from energy conservation applied to a discrete node.
- Interpret the Laplacian finite difference stencil $(T_{i+1} - 2T_i + T_{i-1})$ as a measure of local curvature or departure from the mean of neighbours.
- Explain the role of heat production A_i in modifying the temperature update.
- Connect the discrete finite difference equation to the continuous partial differential equation it approximates.

Consider a vertical column of sediment divided into small layers. Heat flows between layers, and each layer can gain or lose energy depending on its neighbours and any internal heat production.

Rather than solving a continuous equation, we will track how heat moves between *discrete points*.

Part A: Physical intuition

- If a layer is hotter than the ones above and below it, does it gain or lose heat?
- If heat is being produced inside the layer, what happens over time?

Part B: Setting up the system

- Sketch a one-dimensional grid of nodes labeled T_{i-1}, T_i, and T_{i+1}.
- Indicate the spacing Δz.

Part C: Heat flux

Heat flows according to Fourier's law:

$$q_z = -k\frac{\partial T}{\partial z} \tag{1}$$

Tasks:

- Write an expression for heat flowing *into* node i.
- Write an expression for heat flowing *out of* node i.

Part D: Energy conservation

Energy conservation for a small volume gives:

$$\text{Rate of change} = \text{in} - \text{out} + \text{produced}$$

Task: Combine your expressions to derive:

$$\frac{\partial T_i}{\partial t} = \kappa\frac{T_{i+1} - 2T_i + T_{i-1}}{\Delta z^2} + \frac{A_i}{\rho c_p} \tag{2}$$

Part E: Interpretation

Explain in words:

- What does the term $T_{i+1} - 2T_i + T_{i-1}$ represent?
- What role does heat production A_i play?
- What happens if $A_i = 0$?

Key Ideas

Finite difference equations are not numerical tricks—they are simply energy conservation applied locally.

Finite differences are discretised energy conservation. The finite difference heat equation is not an approximation introduced for computational convenience — it is the exact energy balance for a discrete rock volume, with the continuous derivatives replaced by differences between neighbouring nodes. This is the same physics as the heat equation; only the representation changes.

The Laplacian stencil measures curvature. The term $T_{i+1} - 2T_i + T_{i-1}$ is the discrete second derivative of temperature. It is positive when a node is cooler than the average of its neighbours (it receives net heat) and negative when it is warmer (it loses heat). This is exactly how diffusion works: it smooths by removing curvature.

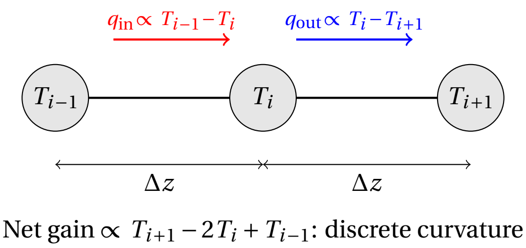


Figure 1: Heat enters node i from the left (red) and exits to the right (blue). The net heat gain is proportional to the curvature of the temperature profile.

The method scales to any geometry. Unlike analytic solutions, which require special symmetries, the same finite difference update rule applies to any grid, any set of boundary conditions, and any spatially variable material properties. This generality is the key advantage of numerical methods.

Activity 10.B
Solving the Heat Equation as a Human Computer

Purpose

To develop intuition for finite difference methods by acting as the nodes in a numerical heat flow model.

In a numerical model, each point (node) updates its temperature using only information from its neighbours.

In this activity, you will act as the nodes in a finite difference model.

Learning Outcomes

By the end of this activity, you should be able to:

- Apply the finite difference update rule by computing temperature from neighbouring values.
- Predict qualitatively how an initial temperature distribution evolves under the diffusion update rule.
- Explain why temperature peaks smooth out and why the system approaches a uniform (or linear) steady state.
- Describe why heat transfer is a local process and how information about temperature propagates through a grid one node at a time.

Box .1: Human Computer

Before electronic computers existed, a *computer* was literally a person employed to carry out calculations by hand or with mechanical calculators. Large groups of human computers were used in astronomy, navigation, ballistics, engineering, and later for the numerical solution of differential equations. By the late 19th and early 20th centuries, many of these computing groups were composed largely of women. Examples include the astronomical computers working for the *Nautical Almanac* and later teams at observatories, government laboratories, and research institutions.

During World War II, the scale of numerical work *exploded* (pun intended). Hundreds of human computers computed ballistic firing tables and solved systems of finite-difference equations for artillery trajectories, fluid flow, and other military problems. Post-WWII, human computers were also instrumental to the US space program. If you've seen the movie *Hidden Figures*, you learned about some of these remarkable women who helped make going to the Moon possible by producing many of the calculations needed to determine flight plans, rocket burn parameters, launch windows, and safe re-entry trajectories.

Finite-difference methods were especially important because many physical problems could not be solved analytically. Human computers discretized differential equations onto grids and repeatedly computed numerical updates row by row—an extremely laborious process that helped motivate the development of electronic computers. The first six programmers of the world's first general-purpose electronic computer, ENIAC, were women who had previously worked as human computers.

David Alan Grier, **2005**, *When Computers Were Human*, Princeton University Press.
Kathy Kleiman, **2022**, *Proving Ground: The Untold Story of the Six Women Who Programmed the World's First Modern Computer*.
David Alan Grier, **2001**, *Human computers: the first pioneers of the information age*, **Endeavour** 25(1).

Setup

- Each student is assigned a position in a line.
- Each student is given an initial temperature value T_i.
- The two students at the ends of the line represent **fixed boundary conditions**.

Update rule

At each step, interior nodes update their temperature according to:

$$T_i^{\text{new}} = \frac{T_{i-1} + T_{i+1}}{2} \tag{3}$$

This means:

- you look at your two neighbours,
- take the average,
- and update your value.

Question 1. Prediction
Before starting, discuss:

- What happens to a sharp temperature spike over time?
- Will the system become smoother or less smooth?

Question 2. Iteration
Repeat the following for several steps:

1. Show your value to your neighbours.
2. Compute your new value.
3. Update simultaneously.

Record the evolution of the system. (Use the table below to track temperature values at each node across three time steps.)

-

Node \ Step	1	2	3	4	5	6	7	8	9	10
0										
1										
2										
3										
4										
5										
6										
7										
8										

Question 3. Observations
Discuss:

- Where do the largest changes occur?
- What happens to temperature peaks?
- How does heat move through the system?

Question 4. Interpretation

Explain in words:

- Why does the system become smoother over time?
- What role do neighbouring nodes play?
- How does this relate to diffusion?

Optional Extension 1: Heat Production

Introduce a constant heat source at one node.

New update rule:

$$T_i^{\text{new}} = \frac{T_{i-1} + T_{i+1}}{2} + S \tag{4}$$

where S is a constant heat input.

Tasks:

- What happens to the system over time?
- Does it still become uniform?
- What steady-state behaviour emerges?

Optional Extension 2: Two-Dimensional Grid

Arrange students in a grid.

Update rule:

$$T_{i,j}^{\text{new}} = \frac{T_{i+1,j} + T_{i-1,j} + T_{i,j+1} + T_{i,j-1}}{4} \tag{5}$$

Tasks:

- Observe how a hot region spreads in two dimensions.
- Compare to the one-dimensional case.
- Describe the shape of the temperature field over time.

Final Reflection

- In what sense are you "solving the heat equation"?
- Why does this method require many iterations?

Key Ideas

Heat flow is a local process: each point evolves based on its neighbours.

Numerical models evolve through local interactions. Each node only communicates with its immediate neighbours. Global behaviour (e.g., a uniform steady state) emerges from many local interactions over many time steps. There is no global solution found in one step.

Information propagates at a finite rate. Because each update depends only on nearest neighbours, a temperature change at one end of the chain cannot instantaneously affect the other end. It must propagate node by node, step by step. This is the discrete analogue of finite diffusion speed in the continuum.

Steady state is reached when updates become zero. The system reaches a steady state when every node's temperature equals the average of its neighbours: $T_i = (T_{i-1} + T_{i+1})/2$. This is the discrete form of $\nabla^2 T = 0$ — the Laplace equation for heat without sources.

Activity 10.C
Boundary Conditions and Geological Meaning

Purpose

To understand how boundary conditions encode geological assumptions in numerical models.

A numerical model does not "know" geology—it only knows equations and boundary conditions.

Learning Outcomes

By the end of this activity, you should be able to:

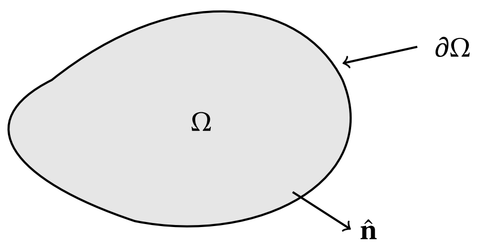
- Distinguish between Dirichlet (fixed value) and Neumann (fixed flux) boundary conditions.
- Match appropriate boundary conditions to geological settings (ocean floor, continental lithosphere, cooling intrusion).
- Explain how changing a boundary condition changes the physical system the model represents.
- Argue why inappropriate boundary conditions can produce physically misleading results even when the numerical solution is mathematically stable.

Boundary conditions determine what kind of Earth your model represents.

Boundary Conditions

A partial differential equation (PDE) describes how a field varies *inside* a domain. To obtain a unique solution, we must also specify how the domain interacts with its surroundings. This additional information is supplied through **boundary conditions**.

Consider a potential variable $u(\mathbf{x})$ defined within a domain Ω bounded by $\partial\Omega$.



The PDE determines the behaviour of the field within Ω. The boundary conditions describe what happens along $\partial\Omega$.

Dirichlet Boundary Condition

$$u = f(\mathbf{x}) \qquad \text{on } \partial\Omega$$

A Dirichlet boundary condition specifies the *value* of the field at the boundary.

Physical interpretation: The environment outside the model imposes a known value on the boundary.

What is fixed?

$$\boxed{\text{Field value}}$$

Neumann Boundary Condition

$$\frac{\partial u}{\partial \hat{\mathbf{n}}} = g(\mathbf{x}) \qquad \text{on } \partial\Omega$$

where n is the outward normal direction.

For heat flow, this is often written as

$$-k\frac{\partial T}{\partial n} = q.$$

A Neumann boundary condition specifies the *flux* crossing the boundary.

Physical interpretation: Rather than fixing the field itself, we prescribe the rate at which energy, mass, fluid, or current enters or leaves the domain.

Examples:

- Mantle heat flux entering the base of the lithosphere.
- Groundwater recharge entering an aquifer.
- Prescribed electrical current density in an electrical model.

What is fixed?

$$\boxed{\text{Flux}}$$

Robin Boundary Condition

$$a(\mathbf{x})u + b(\mathbf{x})\frac{\partial u}{\partial \hat{\mathbf{n}}} = c(\mathbf{x}) \qquad \text{on } \partial\Omega$$

A Robin boundary condition combines a field value and a flux condition.

For example,

$$-k\frac{\partial T}{\partial n} = h(T - T_{\text{atm}})$$

describes heat exchange between a surface and the atmosphere.

Physical interpretation: The boundary interacts with an external environment. Neither the field value nor the flux is fixed independently; instead they are linked together.

Examples:

- Heat exchange between rock and the atmosphere.
- Cooling of a magmatic intrusion by surrounding country rock.
- Groundwater leakage through a semi-permeable boundary.

What is fixed?

$$\boxed{\text{A relationship between value and flux}}$$

Cauchy Boundary Condition

$$u = f(\mathbf{x}), \qquad \frac{\partial u}{\partial \hat{\mathbf{n}}} = g(\mathbf{x}) \qquad \text{on } \partial\Omega$$

A Cauchy boundary condition specifies both the field value and its normal derivative.

Physical interpretation: The boundary is strongly constrained by observations.

Examples:

- A borehole where both temperature and heat flow are measured.
- A boundary constrained simultaneously by observed hydraulic head and groundwater flux.
- Inverse problems where multiple independent observations are available along a boundary.

What is fixed?

$$\boxed{\text{Value and flux}}$$

Summary

Boundary Condition	Prescribes	Physical Meaning
Dirichlet	Field value	Known state
Neumann	Flux	Known transfer rate
Robin	Value + flux relation	Exchange with surroundings
Cauchy	Value and flux	Strong observational constraint

Key Idea The governing equation describes the physics occurring within the model domain. Boundary conditions describe how the model interacts with the outside world. Two models can use exactly the same PDE and physical properties yet produce completely different results if different boundary conditions are chosen. Choosing boundary conditions is therefore a geological modelling decision, not merely a mathematical one.

Part A: Grounding in real systems

Consider the following settings:

- Cooling oceanic crust
- Stable continental lithosphere
- A cooling granite pluton
- Antarctic ice sheet geothermal model

Task: For each case, describe what controls the temperature at the boundaries.

Part B: Types of boundary conditions

Common thermal boundary conditions:

- Fixed temperature (Dirichlet)
- Fixed heat flux (Neumann)

Task: Match each type of boundary condition to a geological situation.

Part C: Comparing assumptions

Explain how the model changes if you prescribe:

- surface temperature
- surface heat flow

What is fixed in each case? What is allowed to vary?

Part D: Geological consequences

Compare appropriate boundary conditions for:

- oceanic lithosphere
- continental lithosphere

Part E: Critical thinking

Task: Explain why inappropriate boundary conditions can produce physically misleading results—even if the numerical solution is mathematically stable.

Reflection
If two models use the same equation but different boundary conditions, can they produce completely different results? Why?

Demo

There is a simple python GUI, `finitediff_gui.py`, that you can use to explore calculations similar to the ones we just did in this exercise. The implementation uses the discrete equation

$$T_i^{n+1} = \frac{k_{i-1}T_{i-1}^n + k_{i+1}T_{i+1}^n}{k_{i-1} + k_{i+1}} + \frac{A_i}{k_i}$$

Key Ideas

Boundary conditions are where geological assumptions enter a numerical model.

Boundary conditions define the physical system, not just the mathematics. Two models with identical equations but different boundary conditions represent completely different geological systems. Choosing a fixed temperature surface is equivalent to assuming an ocean floor in contact with circulating seawater; choosing a fixed heat flux is equivalent to assuming a continental surface where temperatures are free to evolve.

Dirichlet and Neumann conditions represent different physical constraints. A Dirichlet condition fixes the value (temperature); a Neumann condition fixes the gradient (heat flux through the surface). The oceanic upper boundary is naturally Dirichlet; the lower lithospheric boundary is often Neumann (fixed mantle heat flux into the base).

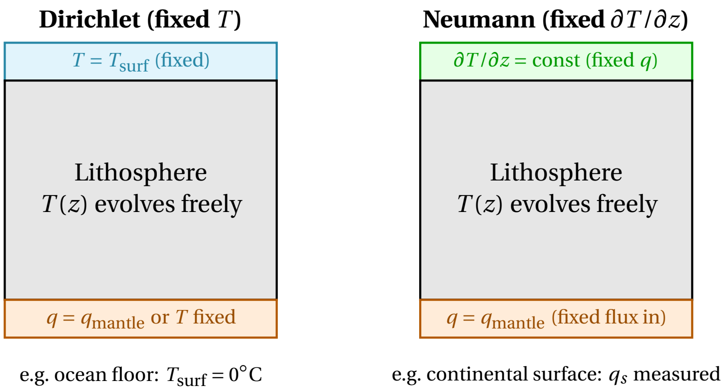


Figure 2: Dirichlet (left) fixes temperature at a boundary — appropriate where temperature is controlled externally (ocean floor). Neumann (right) fixes the heat flux gradient — appropriate where a flux is imposed but temperature is free to adjust (base of lithosphere, continental surface heat flow).

Geological understanding is required to choose appropriate boundary conditions. A numerically stable model with inappropriate boundary conditions can produce results that look reasonable but represent geology that does not exist. Interpreting model output always requires checking that the boundary conditions faithfully represent the geological setting being modelled.

Activity 10.D
Stability, Time Stepping, and Information Transfer

Purpose

To develop physical intuition for numerical stability in explicit finite difference schemes.

In a time-stepping model, temperature is updated step-by-step. Each step uses information from neighbouring nodes.

Think of heat as spreading gradually through the system.

Learning Outcomes

By the end of this activity, you should be able to:

- State the stability condition for the explicit finite difference heat equation and identify the quantities it depends on.
- Explain physically why the allowable time step is proportional to Δz^2 and inversely proportional to κ.
- Describe what physically happens when the stability limit is exceeded, and why this corresponds to physically impossible behaviour.
- Compare stability requirements for materials with different thermal diffusivities.

Part A: Prediction

Consider two cases:

- very small time steps
- very large time steps

What do you expect to happen in each case?

Part B: Physical reasoning

Task: Explain why heat cannot move arbitrarily far in a single time step.

What determines how quickly heat diffuses through a material?

Part C: Stability condition

For a one-dimensional explicit scheme:

$$\Delta t < \frac{\Delta z^2}{2\kappa} \tag{6}$$

Tasks:

- Why does the allowable time step depend on Δz?

- Why does it depend on thermal diffusivity κ?

Part D: Interpretation

Task: Interpret the stability condition as a limit on how fast thermal information can move between nodes.

Why can heat not "jump" over nodes in a single time step?

Part E: Geological interpretation

Explain:

- Why are sedimentary basins (low κ) easier to model?

- Why do high-diffusivity materials require smaller time steps?

Key Ideas

Numerical instability occurs when the model allows heat to propagate faster than physically possible.

The stability condition is a physical causality constraint. Heat diffuses at a rate set by κ. In a time step Δt, physical heat can travel at most $\sqrt{\kappa \Delta t}$. The stability condition $\Delta t < \Delta z^2/2\kappa$ ensures that this thermal length cannot exceed the node spacing Δz. Violating the condition means asking the model to move heat further in one step than the physics allows.

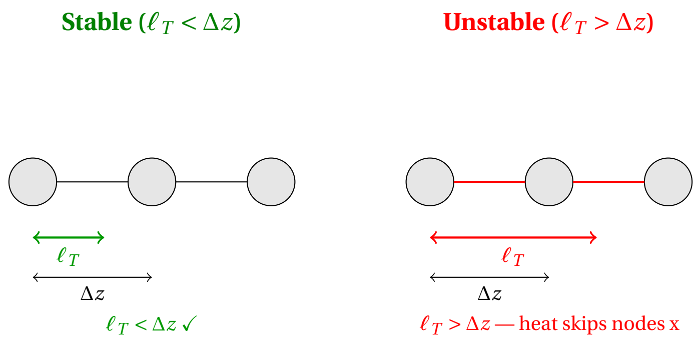


Figure 3: Stability requires that the thermal length $\ell_T = \sqrt{\kappa \Delta t}$ does not exceed the node spacing Δz. When it does, heat is asked to propagate past neighbouring nodes in one step, which is physically impossible and causes numerical blow-up.

Finer grids require smaller time steps. Because stability scales as Δz^2, halving the grid spacing reduces the maximum allowable time step by a factor of four. Increasing resolution is therefore expensive: the number of spatial nodes doubles *and* the number of time steps quadruples to cover the same total time.

High diffusivity is more demanding numerically. Materials with large κ (e.g., peridotite, ice) diffuse heat quickly and require smaller time steps for numerical stability than low-diffusivity materials (e.g., clay-rich sediments). This is the reverse of the geological situation: fast-diffusing materials are easier to model geologically but harder numerically.

Activity 10.E
Verification Against Analytic Solutions

Purpose

To demonstrate the importance of verifying numerical models against solutions with known behaviour.

Learning Outcomes

By the end of this activity, you should be able to:

- Identify analytic benchmark solutions appropriate for testing a numerical heat flow implementation.
- Describe which properties of a numerical solution should be compared with the analytic result (profile shape, heat flow, temporal evolution).
- Explain how grid spacing and time-step size affect numerical accuracy and distinguish accuracy from stability.
- Justify why verification is essential before using a numerical model for geological prediction.

Analytic solutions provide benchmarks for testing numerical implementations.

Task 1: Choose a benchmark

Identify one analytic solution from earlier in the course that could be reproduced with the finite difference code (e.g., linear steady-state geotherm, half-space cooling, Fourier series plate cooling).

Describe the solution and the conditions it requires.

Task 2: What to compare

Describe which aspects of the solution you would compare between the analytic result and the numerical output.

Consider: temperature profile shape, heat flow at the surface or base, behaviour at late times.

Task 3: Grid and time-step effects

Explain how decreasing grid spacing Δz and time step Δt affects the accuracy of the numerical solution.

What limits how small you can make these in practice?

Task 4: Stable but wrong

Explain why a numerically *stable* solution is not necessarily *accurate*.

Give an example of a stable but inaccurate result.

Reflection

Why is verification especially important when modelling complex geological systems where no field observation can directly confirm the model output?

Key Ideas

Verification is not the same as validation. Verification checks that the code correctly solves the equations as written (by comparing to an analytic solution). Validation checks that the equations represent the real system. Both are necessary, and a model can pass verification while still being physically wrong.

Convergence with decreasing grid size is the key accuracy test. If the numerical solution approaches the analytic solution as $\Delta z \to 0$ and $\Delta t \to 0$, the scheme is convergent. The rate at which it converges (first or second order) indicates how quickly errors decrease with refinement.

Stability and accuracy are independent properties. A scheme is stable if errors do not grow unboundedly; it is accurate if the truncation error is small. Choosing Δt just below the stability limit keeps the solution stable but may leave large truncation errors. Choosing $\Delta t \ll \Delta z^2/2\kappa$ is more accurate but computationally expensive.

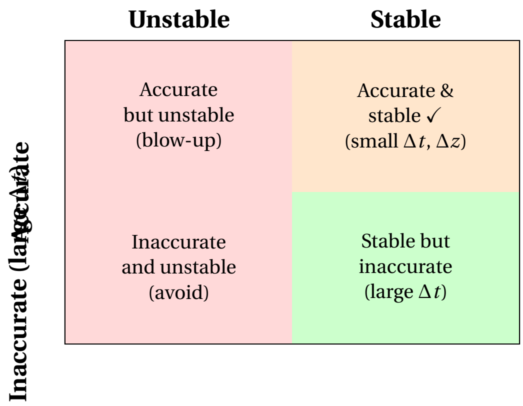


Figure 4: Stability and accuracy are independent. The target is the upper-right quadrant: stable *and* accurate. Stability alone (lower-right) is not sufficient for reliable geological prediction.

Activity 10.F
Building Geological Intuition with Numerical Experiments

Purpose

To use numerical experiments to explore how thermal structure responds to geological complexity.

Learning Outcomes

By the end of this activity, you should be able to:

- Predict qualitatively how depth-dependent conductivity, layered heat production, time-varying basal heat flux, or a transient intrusion would change a model's temperature field.
- Distinguish which geological complexities can still be handled analytically and which require a numerical approach.
- Explain how a numerical model isolates physical effects that cannot be separated observationally.

Consider modifying a numerical heat flow model to include:

- depth-dependent thermal conductivity,
- layered heat production,
- time-varying basal heat flux,
- transient magma intrusions.

Task 1: Qualitative predictions

For each modification listed above, predict how the temperature field would respond compared to a uniform, steady-state reference model.

Task 2: Analytic vs numerical

Identify which of the modifications could still be handled with an analytic solution, and which require a fully numerical approach. Justify your choices.

Task 3: Isolating effects

Explain how a numerical model allows you to separate physical effects that are difficult to disentangle from field observations.

Give one geological example where this ability is particularly valuable.

Key Ideas

Numerical modelling enables controlled experiments on Earth systems that cannot be performed in nature.

Numerical models are controlled experiments. Unlike field observations, numerical experiments allow one variable to be changed at a time while holding all others fixed. This is the only practical way to isolate the thermal effect of, for example, a single high-conductivity layer within a complex lithosphere.

Analytic solutions fail when properties vary in space or time. The steady-state heat equation has a closed-form solution for uniform conductivity and constant heat production. Depth-dependent k, layered A, or a time-varying basal flux break these assumptions and generally require numerical integration.

Qualitative intuition guides model design. Before running a model, predicting the sign and magnitude of an effect (will a high-conductivity layer raise or lower temperatures above it?) builds the physical intuition needed to detect numerical errors and interpret model output.